

🔗 “Explain how to set up and configure a small-to-medium size office network.”

Lesson Title: Network Setup and Initial Configuration

Lesson Objectives / Outcomes

By the end of this lesson, students should be able to:

1. Understand the basic components needed in a small-to-medium office network.
2. Explain the steps to set up and configure network devices.
3. Recognize the importance of proper IP addressing, security, and testing.
4. Perform basic troubleshooting after network setup.

I. Introduction

Every organization, whether small or large, needs a **network** to connect computers, printers, and servers so they can share resources and access the internet efficiently.

Setting up a **small-to-medium office network** involves careful planning, proper configuration, and testing.

II. Basic Network Components

To set up a functioning network, the following components are needed:

Component	Function
Router	Connects the local network to the internet.
Switch	Connects multiple devices within the same network.
Access Point (AP)	Provides wireless (Wi-Fi) access to users.
Modem	Converts internet signal from the ISP.
Cables (UTP Cat5e/Cat6)	Physical medium for wired connections.
Computers, Printers,	End devices that use the network.

Servers

III. Steps to Set Up a Small-Medium Office Network

1. Network Planning

- Determine the **number of devices** (computers, printers, etc.).
- Identify if the network will be **wired, wireless, or both**.
- Assign an **IP addressing scheme** (example: 192.168.10.0/24).
- Decide where to place **routers, switches, and access points** for best coverage.

2. Physical Setup

- Connect the **modem** to the **router's WAN port**.
- Connect the **router's LAN port** to the **switch** using a network cable.
- Connect **computers and printers** to the switch (for wired).
- Install **access points** for wireless connections in strategic areas.

3. Router Configuration

1. Access the router's web interface (e.g., 192.168.1.1).
2. Set up:
 - **Admin password** (for security)
 - **SSID (Wi-Fi name)** and **Wi-Fi password**
 - **DHCP range** (example: 192.168.10.10 to 192.168.10.100)
 - **DNS and gateway settings**
3. Save and restart the router.

4. Switch Configuration (if managed)

- Assign a management IP (example: 192.168.10.2).

- Configure **VLANs** (if separating departments or guests).
- Connect all devices to their assigned VLAN ports.

5. Device Configuration

- Ensure all devices obtain IP addresses automatically (DHCP) or assign static IPs for critical equipment like servers or printers.
- Check network connectivity using the **ping** command.

IV. Network Security Setup

To keep your office network safe:

1. **Change default router credentials.**
2. **Enable WPA3/WPA2 encryption** for Wi-Fi.
3. **Turn on firewall protection** on the router.
4. **Create a guest Wi-Fi network** (separate from main network).
5. **Update firmware** regularly.

V. Testing and Verification

After configuration:

- Check each device's internet access.
- Test file sharing or printer access.
- Use **ping** and **ipconfig** to confirm proper IP assignment.
- Document network settings and passwords securely.

VI. Troubleshooting Tips

Issue	Possible Cause	Solution
No internet	Cable disconnected or wrong	Check connections, restart router

settings

IP conflict	Duplicate IPs	Enable DHCP or assign unique static IPs
Slow Wi-Fi	Interference or distance	Reposition AP, reduce obstacles
Printer not found	Wrong IP or driver	Check printer's IP and driver setup

VII. Summary

- A small-medium office network connects users and devices to share resources efficiently.
- Main steps: **Plan → Connect → Configure → Secure → Test.**
- Proper configuration of routers, switches, and devices ensures stability and performance.
- Documentation and regular maintenance keep the network running smoothly.

VIII. Short Activity (15–20 minutes)

Instruction:

Each student must create a **simple network diagram** (on paper or digital) showing a small office setup with:

- 1 Router
- 1 Switch
- 1 Access Point
- 3 Computers
- 1 Printer

Label all IP addresses and connections.